Rotated Spherical Grid

Jian-Guo Li

Met Office, Exeter, EX1 3PB, UK
Email: Jian-Guo.Li@metoffice.gov.uk

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1. Rotated latitude-longitude system

We live on an almost perfect sphere, the Earth, and a lot of our mundane events are related to locations on the Earth sphere. The global spherical coordinate system is the most used, which manifests as our well-known latitude and longitude system. Many environment applications are based on the latitude-longitude system, and a lot of numerical models use the latitude-longitude grid. One known problem with the spherical system is its polar problem, which has a singular point at each of the two poles. Our direction is usually defined by a reference to the local east, but it loses its meaning at the poles. Every direction from the North Pole is southward and wherever you go from the South Pole is towards the north. The polar singularity is a pure mathematical problem and sometimes we would like to avoid it by using a different system. Rotated spherical system is a simple one we could use to move the poles out of our concerned area, such as the Arctic. For instance, if we move the new pole to the standard Equator, the Arctic will be in the new equator area without the singular pole. For this reason, the rotated grid is also called the equatorial grid sometimes. Rotated grid has many other useful applications, such as mapping or simplifying some calculations.

Rotated grid is obtained by rotating the North Pole to the new pole position at latitude φ_p and longitude λ_p in the standard latitude-longitude system as shown in Fig. 1. The zero-meridian (zero longitude) for the rotated system is chosen to be on the great circle passing both the standard and new poles and, on the side opposite to the new pole side, that is, the standard longitude of the new zero-meridian is

$$\lambda_0 = \lambda_p + 180^\circ$$

By setting λ_p within (-180˚, 180˚), the zero-meridian λ_0 will be within (0, 360˚). For instance, if the new 'North Pole' is moved to latitude $\varphi_p = 60^\circ$ and longitude $\lambda_p = -180^\circ$, the new zero-meridian will overlap with the standard zero-meridian, that is, $\lambda_0 = 0$.

Any point A on the globe can be specified by either its standard lat-lon coordinates (λ, φ) or its rotated or equatorial lat-lon (λ', φ'). The relationship between the two coordinate systems may be derived by spherical geometry.

If we choose the new zero-meridian cross-section as the x-z plane with *z* along the standard North Pole and *x* towards the zero-meridian side, the standard and new pole vectors can then be expressed as

$$\mathbf{N} = (0,0,1), \quad \mathbf{N}' = (-\cos\varphi_p, 0, \sin\varphi_p)$$

The new x and y unit vectors are given by

$$\mathbf{i}' = (\sin\varphi_p, 0, \cos\varphi_p), \quad \mathbf{j}' = \mathbf{j} = (0,1,0)$$

Assuming the standard longitude difference from the new zero-meridian (or our new x-z plane) to the chosen meridian plane of the point A is

$$\gamma = \lambda - \lambda_0 = \lambda - \lambda_p - 180^\circ$$

where λ is also within (0,360˚) as λ_0 and γ within (-180˚, 180˚), the radial unit vector for point A is then given by

$$\mathbf{A} = \left(\cos\varphi\cos\gamma, \cos\varphi\sin\gamma, \sin\varphi\right)$$

The latitude of the point A in the rotated or equatorial system, φ', is then given by

$$\sin\varphi' = \mathbf{N}'\cdot\mathbf{A} = -\cos\varphi_p\cos\varphi\cos\gamma + \sin\varphi_p\sin\varphi \quad (1)$$

or

$$\varphi' = \arcsin\left(\sin\varphi_p\sin\varphi - \cos\varphi_p\cos\varphi\cos\gamma\right) \quad (1a)$$

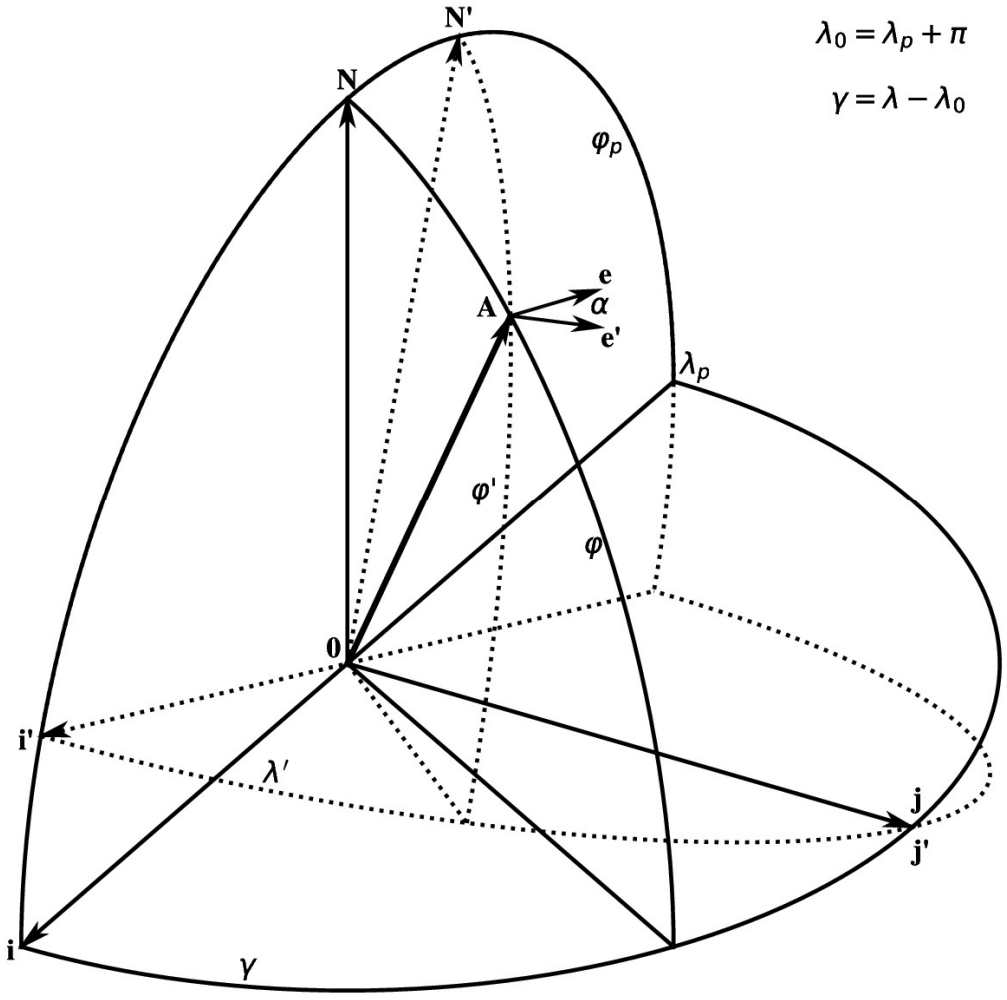


Fig.1. Illustration of the relationship between the standard and rotated latitude-longitude systems.

The longitude of the point A in the rotated system, λ', can be derived from the invariant relationship of the y component as the rotated system shares the same y axis with the standard system:

$$\cos\varphi'\sin\lambda' = \mathbf{j}'\cdot\mathbf{A} = \cos\varphi\sin\gamma \quad (2)$$

If the rotated North Pole is still in the northern hemisphere, the rotated longitude λ' will share the same sign as γ, providing both are within (-180°, 180°). Hence, we have

$$\lambda' = \arcsin\left(\sin\gamma \frac{\cos\varphi}{\cos\varphi'} \right) \tag{2a}$$

If cosφ' = 0, that is, point A is at the new North Pole, λ' may have any value or it is simply chosen as zero. This situation is rare as the aim of the rotated grid is to move the new regional domain to be at middle latitude. Note that arcsin yields a range between $-\pi/2$ and $\pi/2$. Additional information is needed to get the full range longitude in the rotated grid.

Another way to derive the longitude in the rotated system is to use the x' component relation:

$$\cos\varphi'\cos\lambda' = \mathbf{i}' \cdot \mathbf{A} = \cos\varphi_p \sin\varphi + \sin\varphi_p \cos\varphi \cos\gamma \tag{3}$$

If cosφ' is non-zero, that is, point A is not at the new North Pole, λ' is then given by

$$\lambda' = sign(\gamma) \arccos\left(\frac{\cos\varphi_p \sin\varphi + \sin\varphi_p \cos\varphi \cos\gamma}{\cos\varphi'} \right)$$

As arcos returns value from 0 to π, this formula gives the full range longitude in the rotated grid. The rotated longitude shares the same sign as γ. The sign transfer function *sign*(γ) is used in the rotated grid subroutine to make the λ' take the same sign as γ.

2. Transform of standard surface vector to equatorial system

A fixed physical quantity like the horizontal wind velocity would not change but it looks different from different coordinate systems. As shown in Fig. 1, if the wind direction is along the local east in the standard system, it looks rotated by an angle, α, if it is measured from the rotated local east. Using the same unit vectors defined in Fig. 1 the local east unit vectors at point A in the two systems can then be represented as:

$$\mathbf{e} = -\mathbf{i}\sin\gamma + \mathbf{j}\cos\gamma, \quad \mathbf{e}' = -\mathbf{i}'\sin\lambda' + \mathbf{j}'\cos\lambda'$$

The rotated unit vectors can be represented in terms of the standard fixed vectors in the vector form:

$$\mathbf{i}' = \mathbf{i}\sin\varphi_p + \mathbf{N}\cos\varphi_p, \quad \mathbf{j}' = \mathbf{j}$$

Substituting them into the eastern unit vector for the rotated system, we have

$$\mathbf{e}' = -\mathbf{i}\sin\lambda'\sin\varphi_p + \mathbf{j}\cos\lambda' - \mathbf{N}\sin\lambda'\cos\varphi_p$$

One of the rotation coefficients is then given by

$$\cos\alpha = \mathbf{e}\cdot\mathbf{e}' = \sin\varphi_p \sin\gamma \sin\lambda' + \cos\gamma\cos\lambda' \tag{4}$$

Inserting the new Cardician system *x*- and *y*-component equations (2 and 3) from the previous section and rearranging, it becomes

$$\cos\alpha\cos\varphi' = \sin\varphi_p\cos\varphi + \cos\varphi_p\cos\gamma\sin\varphi \tag{4a}$$

The local northern unit vector at point A in the standard system is given by

$$\mathbf{n} = \mathbf{N}\cos\varphi - \sin\varphi\left(\mathbf{i}\cos\gamma + \mathbf{j}\sin\gamma\right)$$

Because **e**' is rotated clockwise from **e**, the angle between **n** and **e**' is $\alpha + \pi/2$. Here the angle α is assumed positive if it is anti-clockwise from the rotated **e**' to the local east **e**. Hence the second rotation coefficient is given by:

$$\sin\alpha = -\mathbf{n}\cdot\mathbf{e}' = -\sin\varphi_p \sin\varphi \sin\lambda' \cos\gamma + \sin\varphi \cos\lambda' \sin\gamma + \cos\varphi_p \cos\varphi \sin\lambda' \tag{5}$$

Similarly inserting the new Cardician system *x*- and *y*-component equations (2 and 3) from the previous section and rearranging, it becomes

$$\sin\alpha\cos\varphi' = \cos\varphi_p \sin\gamma \tag{5a}$$

The rotation angle can be determined by the two rotation coefficients. If a horizontal wind velocity in the standard grid is represented by the eastern and northern components *u* and *v*, their corresponding components in the rotated system (*u'* and *v'*) should be turned anticlockwise by angle of α as the rotated eastern unit vector **e'** is turned clockwise by the angle α from the standard eastern vector.

$$\begin{pmatrix} u' \\ v' \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} \tag{6}$$

Output vectors from the rotated system must be turned clockwise by the same angle to restore the true orientation in the standard system. The reverse conversion of rotated wind into standard system could be derived by using the reciprocal matrix of the above rotation matrix:

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} u' \\ v' \end{pmatrix} \tag{7}$$

The reverse wind conversion can be used to define a solid rotation velocity field on a sphere around a given rotation axis, which will be used for test of advection schemes on a spherical surface grid. The solid sphere rotation has only one eastern velocity component in the rotated grid with the rotation axis to be the new pole, that is, $u' = \omega r\cos\varphi'$ where ω is the angular rotation speed and r is the radius of the sphere, the corresponding wind components in the standard system is then given by

$$u = u'\cos\alpha, \quad v = u'\sin\alpha$$

Inserting $u' = \omega r\cos\varphi'$ and the rotation coefficient expressions, the solid rotation wind components become

$$u = \omega r\left(\sin\varphi_p\cos\varphi + \cos\varphi_p\sin\varphi\cos\gamma\right), \quad v = \omega r\cos\varphi_p\sin\gamma$$

Note $\gamma = \lambda - \lambda_p - \pi$ the solid rotation wind compoents can be rewritten in standard latitude and longitude as

$$u = \omega r\left(\sin\varphi_p\cos\varphi - \cos\varphi_p\sin\varphi\cos\left(\lambda - \lambda_p\right)\right),$$

$$v = \omega r\cos\varphi_p\sin(\lambda - \lambda_p)$$

For wave model application in the polar region, the rotated grid of $\varphi_P = 0$ and $\lambda_P = 180º$ is used to provide the fixed map-east direction for wave spectral definition. The map-east is approximated by the rotated local east. The wind rotation angle α is then used to convert the wave spectra between the map-east and local east direction systems. As the rotated latitude and longitude are not used, a single equation to find out the rotation angle will be enough for wave model application. Since $\lambda_P = 180º$ the converted longitude, $\gamma = \lambda$, is equal to unconverted longitude. Inserting $\varphi_P = 0$ and $\lambda_P = 180º$ into the previous equations (1), the rotated latitude becomes $\sin\varphi' = -\cos\varphi\cos\lambda$, which could be used to derive the cosine of rotated latitude:

$$\cos\varphi' = \sqrt{1 - (\cos\varphi\cos\lambda)^2}$$

Because the map-east is used only in the polar region of the standard grid, the rotated latitude φ' will be restricted around the rotated equatorial region, and hence $\cos\varphi' > 0$. Inserting it into the horizontal vector rotation coefficients (4a and 5a), we have

$$\cos\alpha = \frac{\cos\lambda\sin\varphi}{\sqrt{1-\left(\cos\lambda\cos\varphi\right)^2}}, \qquad \sin\alpha = \frac{\sin\lambda}{\sqrt{1-\left(\cos\lambda\cos\varphi\right)^2}}, \tag{8}$$

Although the sine function is simpler, it is not practical to retrieve the full range rotation angle because arcsine usually returns value from $-\pi/2$ to $\pi/2$. The arccosine function, however, returns value from 0 to π but makes the sign ambiguous. So it is better to use A = arccos(cosα) to derive the angle absolute value and the first equation in (7) to determine the sign because α shares the same sign as sinλ'.

$$\alpha = \mathrm{sgn}(\cos\varphi \sin\lambda) \arccos\left[\frac{\cos\lambda \sin\varphi}{\sqrt{1-(\cos\lambda\cos\varphi)^2}}\right] \tag{9}$$

The rotation angle becomes ambiguous at the North Pole because cos φ = 0 and λ is not defined. Because the map-east coincides with the local east on the zero-meridian, the rotation angle for the North Pole should be chosen as zero, or α = 0. If the longitude at the North Pole is set as λ = 0, and the function sgn returns a positive sign for a zero value (this is the case in FORTRAN but may not be true in other languages, such as IDL), Eq. (9) can produce the desired value α = 0. It is safe to set α = 0 for the North Pole if you are not sure of the subtle difference of the sign function in different languages. Eq. (9) can be verified with the rotation angles of any points close to the Pole along four meridians: λ = 0º, 90º, 180º and -90º. The rotation angle should be approximately α = 0º, 90º, ±180º, and -90º, respectively.

For Arctic wave propagation, the map-east is used within the Arctic region for definition of the wave spectrum. Hence, the Arctic spectrum must be rotated clockwise by the angle α before it is used as a local-east reference spectrum (in the global part northern boundary). That is, the spectral component at angle θ' reference to the map-east becomes the spectral component of angle θ = θ' – α in relative to the local east. Similarly, the global spectrum at angle θ from the local east becomes the spectral component of angle θ' = θ + α in the map-east system (for Arctic part boundary condition).

3. Reversed transformation from equatorial to standard system

For reverse translation from equatorial latitude-longitude to standard latitude-longitude, the above formulations may be used as if the rotated pole is at the same latitude $\varphi_p' = \varphi_p$ and longitude $\lambda'_{p=} 0^\circ$. The transform will then convert the rotated lat-lon to standard lat-lon except that the longitude is measured from the new zero-meridian of the standard system at $\lambda'_0 = 180^\circ$, which is the original rotated pole longitude λ_p in the standard system. Simply add the original rotated pole longitude will complete the transformation. So the reverse formulations are obtained by the following substitutions in (1-4):

$$\gamma \to \lambda' - 180^\circ, \qquad \lambda' \to \lambda - \lambda_p$$
$$\varphi \to \varphi', \qquad \varphi' \to \varphi$$

Hence, we have

$$\sin\varphi = \sin\varphi_p \sin\varphi' + \cos\varphi_p \cos\varphi' \cos\lambda' \tag{10}$$

$$\cos\varphi \sin(\lambda - \lambda_p) = -\sin\lambda' \cos\varphi' \tag{11}$$

The sign of $\lambda - \lambda_p$ will be opposite to λ' as the new zero-meridian is 180° from the old zero-meridian. Equivalently, multiplying by -1 and noting $\lambda_0 = \lambda_p + 180^\circ$, it becomes

$$\lambda - \lambda_0 = \arcsin\left(\sin\lambda' \frac{\cos\varphi'}{\cos\varphi}\right)$$

The sign of $\lambda - \lambda_0$ will be the same as λ'. If cosφ = 0, that is, at the standard pole, λ may take any value or simply assume to be zero.

The reverse rotation angle for the wind vectors will be just the opposite of the original angle. So simply reverse the sign of α in (4) for conversion of wind velocity from rotated system to the standard system as in (7).

Nair and Machenhauer (2002) introduced a double-vortex deformation test on a sphere for spherical grid validation. The non-divergent spherical deformation flow with two symmetric vortices around rotated poles is defined by the angular speed ω as a function of the rotated latitude, φ', relative to its rotational pole as

$$\omega(\varphi') = \omega_0 \frac{\sinh\beta}{\beta\cosh^3\beta}, \quad \beta \equiv 3\cos\varphi'$$

Here ω_0 is the maximum angular speed when β becomes 0 or at the rotational poles. To avoid division by zero in the computer code, ω is set to ω_0 if $\beta = 0$. The angular speed has a minimum value of about $0.0085\omega_0$ on the rotated equator ($\varphi' = 0$). The deformation flow is like the solid-body rotational flow except that the angular speed varies with the rotated latitude.

Note that the surface rotation speed, $u' = \omega r\cos\varphi'$, is proportional to the product of angular speed and the rotation radius at the sphere surface. It is zero at the rotated poles even if the angular speed has the maximum there because the $\cos\varphi'$ is zero at the rotated poles. The rotation speed increases from both rotated poles, and peaks at

$$\varphi'_{Max} = \pm\arccos\left(\sinh^{-1}(\sqrt{2}/2)/3\right) \approx \pm 77.32^{o}$$

or about 12.68^{o} from the rotational poles. For a given initial distribution $\psi(\lambda',\ \varphi')$ as a function of the rotated coordinates, the exact solution at time t generated by this deformation field is given by $\psi(\lambda' - \omega t,\ \varphi')$. These deformation test functions given at rotated point $(\lambda',\ \varphi')$ can be converted to standard grid, using the reverse conversion formula in this section. Two Python functions, `ll2eqdeg.py` and `eq2lldeg.py`, are available for converting a standard latitude-longitude grid into the rotated or equatorial grid and vice versa.

4. Rotation of wave spectra between standard and equatorial systems

Wave spectra will be rotated the same angle as the wind velocity. As wave spectra are represented as discrete directional arrays in the wave model, the rotation will be divided into a simple re-numbering of directional bins and shifting of wave energy by the fractional angle (remainder of the rotation angle divided by the bin width, hence less than one directional bin width). Wave energy will be assumed to be uniform within one directional bin, so the fractional shift will be simply passing the fractional energy from the 'upstream' bin to the lower one. The equal amount will be deducted from the upstream bin, so the total wave energy is conserved and remain positive-definite (in analogous to the first order upstream advection scheme).

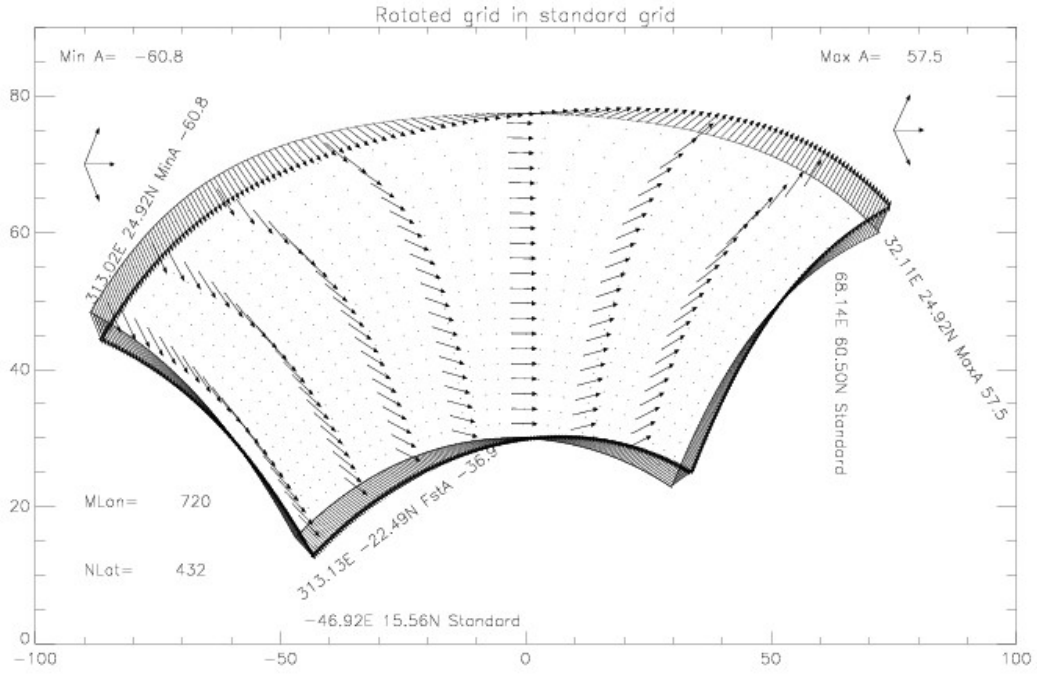


Fig.2. The rotated NAE domain plotted in the standard lat-lon system. The wind velocity arrows are rotated pure westerly wind in the standard system and plotted in reference to the standard system.

The North Atlantic and European (NAE) model is used as an example to demonstrate the rotated grid and the wave spectral rotation. The NAE model domain is set to be in a rotated grid with the new pole at PLAT=37.5, PLON=177.5. The mesh is uniform lat-lon 0.11 deg in the rotated grid with 720 longitude points from 313.02 and 432 latitude points from -22.49. The rotated grid point locations in standard grid are showed Fig.2:

Wave spectra at lateral boundaries from the standard grid model must be rotated before being used as boundary conditions in the rotated grid. A wave spectral rotation program is provided in the WAVEWATCH III model. One spectral rotation example is illustrated in Fig.3. The wave spectral is a typical cosine square distributed JONSWAP spectrum with 25 spectral bins and 24 directions. The peak frequency is about 0.1 Hz and the mean direction is about 45 degree from the north in the northeast (central panel in Fig.3). The spectrum is rotated anti-clockwise by 80 degree and is shown in the left panel in Fig.3. The right panel in Fig.3 compares the 1-D spectra of the original and rotated 2-D spectra. They are identical as expected because rotation does not affect the 1-D spectra.

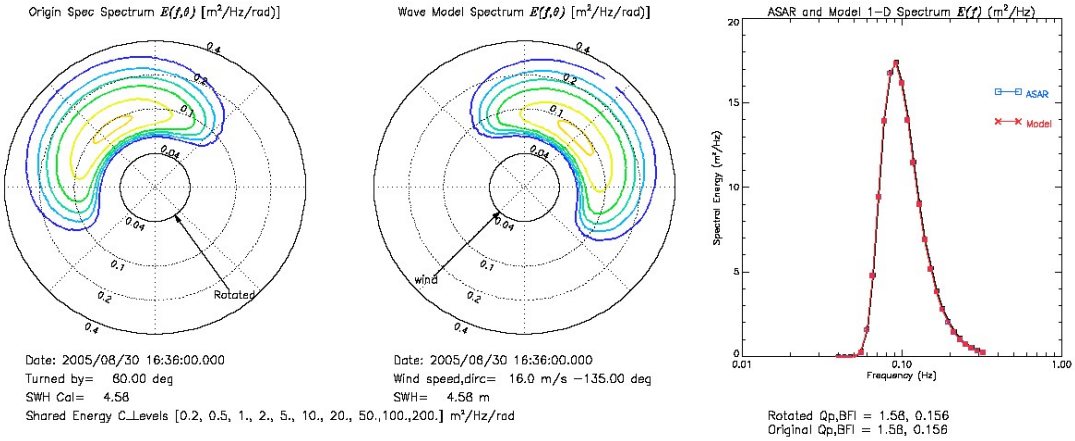


Fig.3. Rotation of wave spectra for rotated system. The original spectrum is shown in the middle panel and the rotated spectrum by the left. The right panel compares the 1-D spectra of the original and rotated 2-D spectra.

5. Stereographic to orthographic projections

Stereographic projection usually means a special mapping approach, which sets the projection point at the South Pole and maps any points on the northern hemisphere on to a plane tangent to the North Pole by extending a straight line crossing the projection point (South Pole) and the mapped point. The Equator circle is projected as a circle with doubled radius. This projection is conformal, that is, it preserves angles at which curves meet, but it is not isometric or does not preserve distances and hence areas. In fact, areas towards the Equator are enlarged more than those close to the North Pole. If the projection point is moved further away from the projection plane to infinite for instance, it becomes the so-called orthographic projection. In this case the projected Equator circle is identical to the real Equator circle of the same radius. However, the orthographic projection shrinks the areas near the Equator more than those near the North Pole. A compromise projection distance is then between the two extremes if we like to keep the Equatorial and polar areas in a better balance. For instance, increasing the distance from the projection plane to 3 times of the sphere radius as shown in Fig. 4, the radius of the projected Equator circle will be reduced from 2 in stereographic projection to 3/2 of the sphere radius, which is closer to the quarter of the circumference ($\pi/2$ times of the spheric radius) than the original stereographic projection. For a general projection point at n times of the sphere radius R from the projection plane, whence this projection will be referred to as the n-radius projection, any point at longitude λ and latitude φ will be projected on the projection plane, tangent to the North Pole, at a distance r from the centre (North Pole) and the ratio of this distance to the radius of the sphere, R, is given by

$$\frac{r}{R} = \frac{n\cos\varphi}{n - 1 + \sin\varphi}$$

The angle from the projected zero-meridian is preserved to be the longitude angle. We could use a rotated North Pole instead of the standard North Pole for projection of any hemisphere centred at the rotated North Pole. Simply converting the standard latitude-longitude $(\varphi,\ \lambda)$ into a rotated one $(\varphi',\ \lambda')$ with a chosen rotated pole position and working out its distance *r* from the rotated pole with the above *n*-radius projection, its position on the projected plane could then be determined with the polar coordinates $(r,\ \lambda')$.

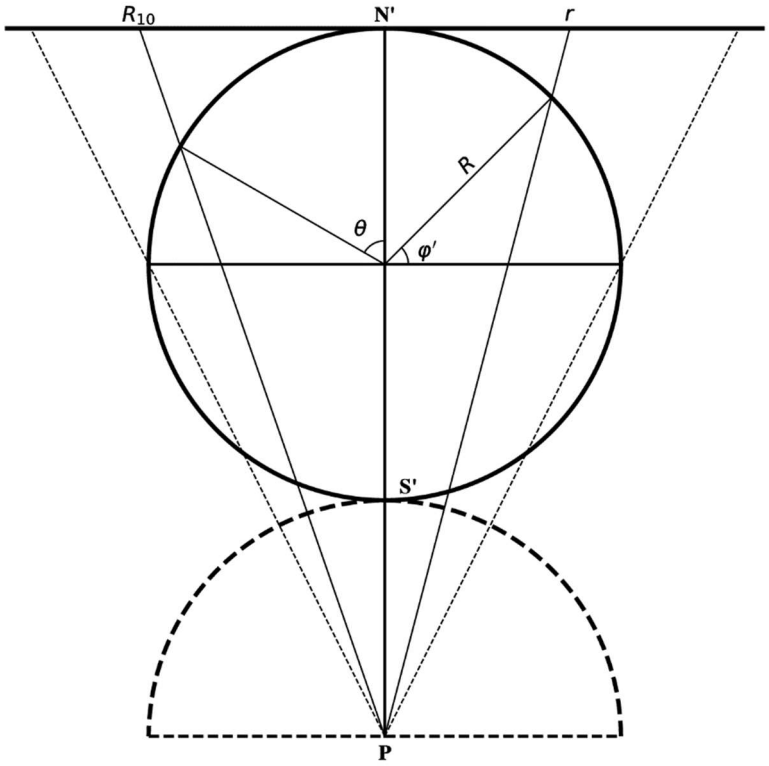


Fig.4. Illustration of the 3-radius projection.

A Python projection function, `steromap.py`, is available for the default 3-radius projection and it also rescales the distance from the rotated pole so that the edge of the rotated parallel at rotated latitude $\varphi' = \pi/2 - \theta$ to be at a default radius of $R_{10} = 10$ units from the rotated north pole N' as shown in Fig.4. The rotated polar area covered within the spanning angle, θ, could then be drawn within a circular area of R_{10} radius. A given point at the rotated location$(\varphi',\ \lambda')$ will be projected on the drawing plane from the rotated pole at a rescaled distance *r*, given by

$$r = R_{10} \cdot \frac{n - 1 + \cos\theta}{n\sin\theta} \cdot \frac{n\cos\varphi'}{n - 1 + \sin\varphi'}$$

and the point coordinates in the projection plane are given by

$$x = -r\sin\lambda', \quad y = r\cos\lambda', \quad for\ \varphi' \geq 0$$

where the rotated zero-meridian is drawn at the top centre of the projection plane for the rotated north hemisphere $(\varphi' > 0)$. If you like the rotated southern hemisphere is drawn, reverse the *x* value to the opposite sign, that is, $x = r\sin\lambda', \quad for\ \varphi' < 0$, which is equivalent to swap the rotated southern hemisphere with the northern one for an equivalent projection because $\cos(-\varphi) = \cos\varphi$. Reversing the *x* sign for the southern hemisphere is just like we view the mirror reflection because the projection point is virtually moved to the northern side at the same distance of *n* times of the radius. The spanning angle θ is a convenient parameter to zoom in the rotated polar region for a local area plot. For a hemi-spherical drawing, simply set the spanning angle to be $\theta = 90^{\circ}$, which is the default value of the spanning angle in the Python program.